

STAT 207: Optimization

- Express the number 10 as a sum of two nonnegative numbers whose product is as large as possible.
- You need to fence in a rectangular play zone for children, to fit into a right - triangular plot with sides measuring 4 m and 12 m. What is the maximum area for this play zone?
- You are a lifeguard on a straight beach. A swimmer is in trouble in the water, what is the minimum time spent for you to reach and rescue the swimmer given the following:
 - Swimmer's location: 100 meters along the beach, 50 meters out into the water.
 - You run faster on sand than you swim in water:
 - Running speed: 6 m/s
 - Swimming speed: 2 m/s

Optimization considers the problem

$$\begin{array}{l} \text{minimize } f(x) \\ \text{subject to constraints on } x \end{array}$$

- Maximization or minimization
- Exact or iterative solutions

Why is optimization important in statistics?

- Parameter Estimation:
 - Maximum likelihood estimation (MLE).
 - Maximum a posteriori (MAP) estimation in Bayesian framework.
- Model Selection:
 - AIC/BIC;
 - LASSO, penalized regression.
- Machine learning:

- Training neural networks involves minimizing a loss function + certain regularizations.

Commonly used optimization methods:

- Newton type algorithms
- Quasi-Newton algorithm
- Expectation-maximization (EM) algorithm
- Majorization-minimization (MM) algorithm
- Gradient Descent

Basic results

Suppose $f(x)$ is differentiable on the open set U :

- differential $df(x)$
- gradient $\nabla f(x)$
- second differential (Hessian) $d^2f(x) = \nabla^2 f(x)$

(Fermat) Suppose a differentiable function $f(x)$ has a local minimum at the point y of the open set U . Then $\nabla f(x)$ vanishes at y .

- Stationary point y : $\nabla f(y) = 0$

NAS Proposition 11.2.3 Suppose a twice continuously differentiable function $f(x)$ has a local minimum at the point y of the open set U . Then $d^2f(x)$ is positive semidefinite at y . Conversely, if y is a stationary point and $d^2f(y)$ is positive definite, then y is a local minimum.

- A function f is coercive if $\lim_{\|x\|_2 \rightarrow \infty} f(x) = \infty$.

Example: $f(x) = \frac{1}{2}x^T A x + b^T x + c$, where A is positive definite.

Example Show that the sample mean and sample variance are the MLE of the theoretical mean and variance of a random sample $y_1, y_2, \dots, y_n$ from a multivariate normal distribution.

Convexity

A function $f : R^n \rightarrow R$ is **convex** if

- $\text{dom} f$ is a convex set: $\alpha x + (1 - \alpha)y \in \text{dom} f$ for all $x, y \in \text{dom} f$ and any $\alpha \in (0, 1)$, and
- $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$ for all $x, y \in \text{dom} f$ and any $\alpha \in (0, 1)$.

f is **strictly convex** if the inequality is strict for all $x \neq y$ and α .

Supporting hyperplane inequality A differentiable function f is convex if and only if

$$f(y) \geq f(x) + df(x)(y - x)$$

for all $x, y \in \text{dom} f$.

Second-order condition for convexity A twice differentiable function f is convex if and only if $\nabla^2 f(x)$ is PSD for all $x \in \text{dom} f$. It is strictly convex if and only if $\nabla^2 f(x)$ is PD for all $x \in \text{dom} f$.

Convexity and global optima

Suppose f is a convex function.

- Any stationary point y , i.e., $\nabla f(y) = 0$, is a global minimum. (Proof: By supporting hyperplane inequality, $f(x) \geq f(y) + \nabla f(y)^\top (x - y) = f(y)$ for all $x \in \text{dom} f$.)
- Any local minimum is a global minimum.
- The set of (global) minima is convex.
- If f is strictly convex, then the global minimum, if exists, is unique.

Example: Least squares estimate. $f(\beta) = \frac{1}{2} \|y - X\beta\|_2^2$ has Hessian $\nabla^2 f = X^\top X$ which is positive semidefinite. So f is convex and any stationary point (solution to the normal equation) is a global minimum. When X is rank deficient, the set of solutions is convex.

Jensen's inequality

If h is convex and W a random vector taking values in $\text{dom} f$, then

$$\mathbb{E}[h(\mathbf{W})] \geq h[\mathbb{E}(\mathbf{W})],$$

provided both expectations exist. For a strictly convex h , equality holds if and only if $W = \mathbf{E}(W)$ almost surely.

Proof: Take $x = W$ and $y = \mathbf{E}(W)$ in the supporting hyperplane inequality.

Information inequality

Let f and g be two densities with respect to a common measure μ and $f, g > 0$ almost everywhere relative to μ . Then

$$\mathbb{E}_f[\log(f)] \geq \mathbb{E}_f[\log(g)],$$

with equality if and only if $f=g$ almost everywhere on μ .

Proof: Apply Jensen's inequality to the convex function $-\ln(t)$ and random variable $W=g(X)/f(X)$ where $X \sim f$.

Important applications of information inequality: M-estimation, EM algorithm.

\vspace{50mm}

Optimization with Equality Constraints

- Suppose the objective function $f(x)$ to be minimized is continuously differentiable and defined on R^n .
- The gradient direction $\nabla f(x) = df(x)^T$ is the direction of steepest ascent of $f(x)$ near the point x .
- The following linear approximation is often used

$$f(x + su) = f(x) + sdf(x)u + o(s),$$

for a unit vector u and a scalar s .

Lagrange multipliers

The Lagrangian function

$$\mathcal{L}(x, w) = f(x) + \sum_{i=1}^m w_i g_i(x),$$

where $f(x)$ is the objective function to minimize, $g_i(x) = 0$ are equality constraints.

```
In [3]: from IPython.display import Image
        Image(filename='gradient.png')
```

Out[3]:

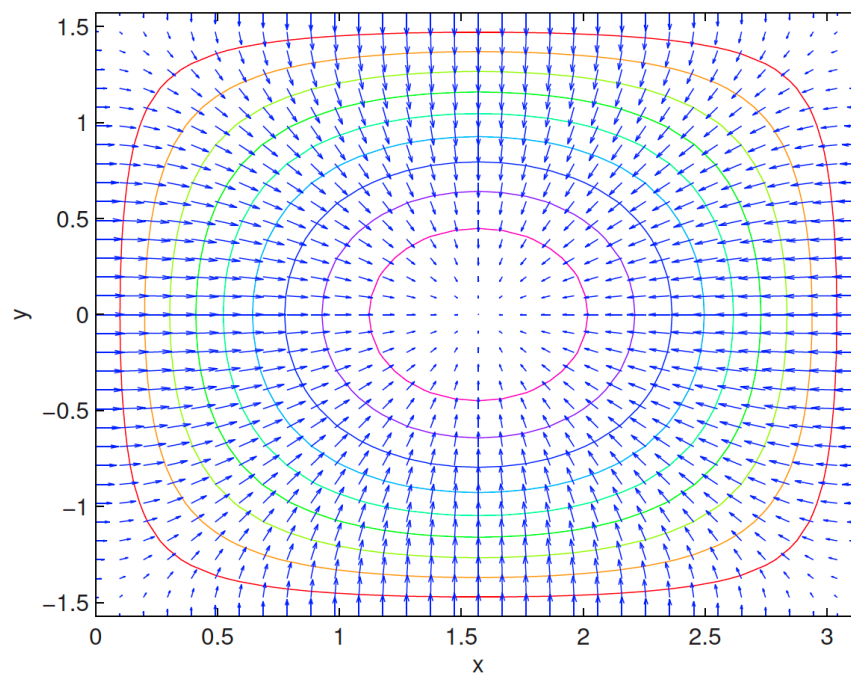


FIGURE 11.1. Level Curves and Steepest Ascent Directions for $\sin(x)\cos(y)$

```
In [ ]: import numpy as np
        import matplotlib.pyplot as plt

        # Define the function
        def f(x, y):
            return np.sin(x) * np.cos(y)

        # Define the range of x and y
        x = np.linspace(-np.pi, np.pi, 100)
        y = np.linspace(-np.pi/2, np.pi/2, 100)

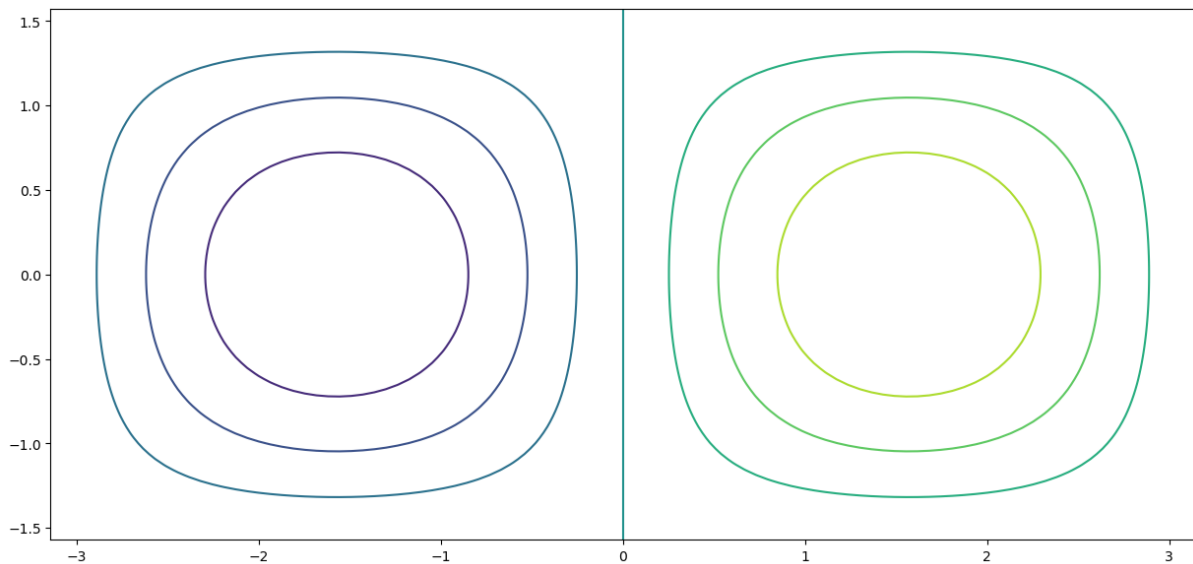
        # Create a grid of (x, y) coordinates
        X, Y = np.meshgrid(x, y)
```

```
# Compute the function values at the grid points
Z = f(X, Y)

# Set up the plot
fig, ax = plt.subplots(figsize=(15, 7))

# Plot the level curves
ax.contour(X, Y, Z)

# Show the plot
plt.show()
```



Proposition 11.3.1 (Lagrange) Suppose the continuously differentiable function $f(x)$ has a local minimum at the feasible point y and that the constraint functions $g_1(x), \dots, g_m(x)$ are continuously differentiable with linearly independent gradient vectors $\nabla g_i(y)$ at y . Then

- there exists a multiplier vector λ such that (y, λ) is a stationary point of the Lagrangian.

Furthermore, if $f(x)$ and all $g_i(x)$ are twice continuously differentiable, then

- $v^T \nabla^2 L(y) v \geq 0$ for every tangent vector v at y .

Conversely, if (y, λ) is a stationary point of the Lagrangian and $v^T \nabla^2 L(y) v > 0$ for every nontrivial tangent vector v at y , then

- y represents a local minimum of $f(x)$ subject to the constraints.

Example Quadratic Programming with Equality Constraints

Minimizing a quadratic function

$$q(x) = \frac{1}{2}x^T Ax + b^T x + c$$

on $\mathbb{R}^n$ subject to the m linear equality constraints

$$v_i^T x = d_i, \quad i = 1, \dots, m$$

is one of the most important problems in nonlinear programming.

To minimize $q(x)$ subject to the constraints, we introduce the Lagrangian

$$L(x, \lambda) = \frac{1}{2}x^T Ax + b^T x + c + \sum_{i=1}^m \lambda_i (v_i^T x - d_i) = \frac{1}{2}x^T Ax + b^T x + c + \lambda^T (V$$

A stationary point of $L(x, \lambda)$ is determined by the equations

$$\begin{aligned} Ax + b + V^T \lambda &= 0, \\ Vx &= d, \end{aligned}$$

whose formal solution amounts to

$$\begin{pmatrix} x \\ \lambda \end{pmatrix} = \begin{pmatrix} A & V^T \\ V & 0 \end{pmatrix}^{-1} \begin{pmatrix} -b \\ d \end{pmatrix}.$$

The inverse exists thanks to the following proposition.

Proposition 11.3.2 Let A be an $n \times n$ positive definite matrix and V be an $m \times n$ matrix. Then the matrix

$$M = \begin{bmatrix} A & V^T \\ V & 0 \end{bmatrix}$$

has inverse

$$M^{-1} = \begin{bmatrix} A^{-1} - A^{-1}V^T(VA^{-1}V^T)^{-1}VA^{-1} & -(VA^{-1}V^T)^{-1}VA^{-1} \\ -A^{-1}V^T(VA^{-1}V^T)^{-1} & (VA^{-1}V^T)^{-1} \end{bmatrix}$$

if and only if V has linearly independent rows $v_{t_1}, \dots, v_{t_m}$.

Additional **Example 11.3.4**

Optimization with Inequality Constraints

Minimize an objective function $f_0(x)$ subject to the mixed constraints

$$h_i(x) = 0, \quad 1 \leq i \leq p,$$

$$f_j(x) \leq 0, \quad 1 \leq j \leq m.$$

A constraint $f_j(x)$ is

- active at the feasible point x provided $f_j(x) = 0$;
- it is inactive if $f_j(x) < 0$.

To avoid redundant constraints, we need

- linear independence of the gradients of the equality constraints,
- and a restriction on the active inequality constraints.

Duality

- The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x}).$$

The vectors $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_m)^\top$ and $\boldsymbol{\nu} = (\nu_1, \dots, \nu_p)^\top$ are called the Lagrange multiplier vectors or dual variables.

- The **Lagrange dual function** is the minimum value of the Lagrangian over $\mathbf{x}$:

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x}} \left(f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x}) \right).$$

- Denote the optimal value of the original problem by p^* . For any $\boldsymbol{\lambda} \succeq 0$ and any $\boldsymbol{\nu}$, we have

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*.$$

Proof: For any feasible point $\tilde{\mathbf{x}}$,

$$\mathcal{L}(\tilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\tilde{\mathbf{x}}) + \sum_{i=1}^m \lambda_i f_i(\tilde{\mathbf{x}}) + \sum_{i=1}^p \nu_i h_i(\tilde{\mathbf{x}}) \leq f_0(\tilde{\mathbf{x}})$$

because the second term is non-positive and the third term is zero. Then,

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \mathcal{L}(\tilde{\mathbf{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f_0(\tilde{\mathbf{x}}).$$

- Since each pair $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ with $\boldsymbol{\lambda} \succeq 0$ gives a lower bound to the optimal value p^*

, it is natural to ask for the best possible lower bound the Lagrange dual function can provide. This leads to the **Lagrange dual problem**

$$\max_{\lambda \geq 0} g(\lambda, \nu),$$

which is a convex problem regardless of whether the primal problem is convex or not.

- We denote the optimal value of the Lagrange dual problem by d^* , which satisfies the weak duality

$$d^* \leq p^*.$$

The difference $p^* - d^*$ is called the optimal duality gap.

- If the primal problem is convex, that is

$$\begin{aligned} \min_{\mathbf{x}} \quad & f_0(\mathbf{x}) \\ \text{subject to} \quad & f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \\ & \mathbf{Ax} = \mathbf{b}, \end{aligned}$$

with $f_0, \dots, f_m$ convex, we usually (but not always) have strong duality, i.e., $d^* = p^*$.

- The conditions under which strong duality holds are called constraint qualifications. A commonly used one is Slater's condition: There exists a point in the relative interior of the domain such that

$$\begin{aligned} f_i(\mathbf{x}) &< 0, \quad i = 1, \dots, m, \\ \mathbf{Ax} &= \mathbf{b}. \end{aligned}$$

Such a point is also called **strictly feasible**.

KKT (Karush, Kuhn, and Tucker) Conditions

- "One of the great triumphs of 20th century applied mathematics."
- Original paper: [Nonlinear Programming by Kuhn and Tucker 1951](#)

Nonconvex problems

- Assume $f_0, \dots, f_m, h_1, \dots, h_p$ are differentiable. Let $\mathbf{x}^*$ and $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ be any primal and dual optimal points with zero duality gap, i.e., strong duality holds.

- Since $\mathbf{x}^*$ minimizes $L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ over $\mathbf{x}$, its gradient vanishes at $\mathbf{x}^*$, we have the Karush-Kuhn-Tucker (KKT) conditions:

1. Stationarity

$$\nabla f_0(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(\mathbf{x}^*) + \sum_{i=1}^p \nu_i^* \nabla h_i(\mathbf{x}^*) = \mathbf{0}.$$

2. Primal feasibility

$$\begin{aligned} f_i(\mathbf{x}^*) &\leq 0, \quad i = 1, \dots, m \\ h_i(\mathbf{x}^*) &= 0, \quad i = 1, \dots, p \end{aligned}$$

3. Dual feasibility

$$\boldsymbol{\lambda}^* \geq \mathbf{0},$$

4. Complementary slackness

$$\boldsymbol{\lambda}^* \cdot \mathbf{f}(\mathbf{x}^*) = 0$$

- The fourth condition (complementary slackness) follows from:

$$f_0(\mathbf{x}^*) = \inf_{\mathbf{x}} \{f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i^* f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i^* h_i(\mathbf{x})\}$$

Since $\sum_{i=1}^m \lambda_i^* f_i(\mathbf{x}^*) = 0$ and each term is non-positive, we have $\lambda_i^* f_i(\mathbf{x}^*) = 0$, $i = 1, \dots, m$.

- To summarize, for any optimization problem with differentiable objective and constraint functions for which strong duality obtains, any pair of primal and dual optimal points must satisfy the KKT conditions.

Convex problems

- When the primal problem is convex, the KKT conditions are also sufficient for the points to be primal and dual optimal.
- If f_i are convex and h_i are affine, and $(\tilde{\mathbf{x}}, \tilde{\boldsymbol{\lambda}}, \tilde{\boldsymbol{\nu}})$ satisfy the KKT conditions, then $\tilde{\mathbf{x}}$ and $(\tilde{\boldsymbol{\lambda}}, \tilde{\boldsymbol{\nu}})$ are primal and dual optimal, with zero duality gap.
- The KKT conditions play an important role in optimization. Many algorithms for convex optimization are conceived as, or can be interpreted as, methods for solving the KKT conditions.

Python Implementations

- [scipy.optimize](#)
- [Duke lectures](#)

In []: